EXPT : 3	EXPLORATORY DATA ANALYSIS (EDA) – DATA CLEANING AND DATA TRANSFORMATION
DATE :	

AIM

To identify and handle missing values, duplicate records, outliers, inconsistent data, and perform data transformation techniques such as normalization and encoding using Python Pandas.

ALGORITHM 1 :

1. Import pandas library
2. Load dataset
3. Use isnull() function
4. Count missing values
5. Display result

Code and Output:

```
import pandas as pd
df=pd.read_csv(r"C:\Users\Keertu\Downloads\tourism_dataset_5000.csv"
)
print(df.isnull())
```

```
      Site Name  Sites Visited  Tour Duration  Route ID  Tourist Rating  \
0         False          False          False     False           False
1         False          False          False     False           False
2         False          False          False     False           False
3         False          False          False     False           False
4         False          False          False     False           False
...         ...            ...            ...       ...             ...
4995      False          False          False     False           False
4996      False          False          False     False           False
4997      False          False          False     False           False
4998      False          False          False     False           False
4999      False          False          False     False           False

      System Response Time  Recommendation Accuracy  VR Experience Quality  \
0                    False                    False                  False
1                    False                    False                  False
2                    False                    False                  False
3                    False                    False                  False
4                    False                    False                  False
...                    ...                      ...                    ...
4995                 False                    False                  False
4996                 False                    False                  False
4997                 False                    False                  False
4998                 False                    False                  False
4999                 False                    False                  False

      Satisfaction
0            False
1            False
2            False
3            False
4            False
...            ...
4995         False
4996         False
4997         False
4998         False
4999         False

[5000 rows x 14 columns]
```

df.isnull().sum()

```
Tourist ID                 0
Age                        0
Interests                  0
Preferred Tour Duration    0
Accessibility              0
Site Name                  0
Sites Visited              0
Tour Duration              0
Route ID                   0
Tourist Rating             0
System Response Time       0
Recommendation Accuracy    0
VR Experience Quality      0
Satisfaction               0
dtype: int64
```

ALGORITHM 2 :

1. Use duplicated()
2. Count duplicate records
3. Remove duplicates

4. Verify dataset

Code and Output :

df.duplicated()

df.drop_duplicates(inplace=True)

print(df)

```
                                          Sites Visited  Tour Duration  \
0     ['Eiffel Tower', 'Great Wall of China', 'Taj M...              7
1                                 ['Great Wall of China']              1
2                                        ['Eiffel Tower']              2
3                           ['Machu Picchu', 'Taj Mahal']              5
4     ['Machu Picchu', 'Taj Mahal', 'Great Wall of C...              7
...                                                 ...            ...
4995                              ['Great Wall of China']              2
4996                                        ['Colosseum']              1
4997                                        ['Taj Mahal']              1
4998 ['Taj Mahal', 'Great Wall of China', 'Colosseum']              7
4999                              ['Great Wall of China']              1

      Route ID  Tourist Rating  System Response Time  Recommendation Accuracy  \
0         1000             1.6                  3.73                       97
1         2000             2.6                  2.89                       90
2         3000             1.7                  2.22                       94
3         4000             2.0                  2.34                       92
4         5000             3.7                  2.00                       96
...        ...             ...                   ...                      ...
4995   4996000             3.9                  2.53                      100
4996   4997000             4.3                  1.53                      100
4997   4998000             3.2                  3.99                       89
4998   4999000             2.5                  2.13                       98
4999   5000000             1.9                  2.99                       92

      VR Experience Quality  Satisfaction
0                       4.5             3
1                       4.5             3
2                       4.7             3
3                       4.7             3
4                       4.8             4
...                     ...           ...
4995                    4.4             4
4996                    5.0             4
4997                    4.4             3
4998                    4.4             3
4999                    4.8             3

[5000 rows x 14 columns]
```

ALGORITHM 3 :

Detecting Outliers

1. Calculate Quartiles
2. Compute IQR
3. Calculate lower and upper limits
4. Identify outliers

Code and Output :

```
Q1 = df['Tourist Rating'].quantile(0.25)
Q3 = df['Tourist Rating'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df['Tourist Rating'] < lower) |
        (df['Tourist Rating'] > upper)]

print(outliers)
```

```
Empty DataFrame
Columns: [Tourist ID, Age, Interests, Preferred Tour Duration, Accessibility, Site Name, Sites Visited, Tour Duration, Route ID, Tourist Rating, System Response Time, Recommendation Accuracy, VR Experience Quality, Satisfaction]
Index: []
```

```
Q1 = df['Tourist Rating'].quantile(0.25)
Q3 = df['Tourist Rating'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower)
print("Upper Bound:", upper)
```

```
Q1: 2.0
Q3: 4.0
IQR: 2.0
Lower Bound: -1.0
Upper Bound: 7.0
```

```
from sklearn.preprocessing import LabelEncoder


encoder = LabelEncoder()


categorical_columns = ['Interests', 'Accessibility', 'Site Name', 'Route ID']


for col in categorical_columns:
    df[col] = encoder.fit_transform(df[col])


print(df)
```

```
      Site Name                                      Sites Visited  \
0             1  ['Eiffel Tower', 'Great Wall of China', 'Taj M...
1             0                            ['Great Wall of China']
2             3                                   ['Eiffel Tower']
3             0                      ['Machu Picchu', 'Taj Mahal']
4             0  ['Machu Picchu', 'Taj Mahal', 'Great Wall of C...
...         ...                                                ...
4995          2                            ['Great Wall of China']
4996          2                                     ['Colosseum']
4997          3                                     ['Taj Mahal']
4998          0  ['Taj Mahal', 'Great Wall of China', 'Colosseum']
4999          0                            ['Great Wall of China']

      Tour Duration  Route ID  Tourist Rating  System Response Time  \
0                 7         0             1.6                  3.73
1                 1         1             2.6                  2.89
2                 2         2             1.7                  2.22
3                 5         3             2.0                  2.34
4                 7         4             3.7                  2.00
...             ...       ...             ...                   ...
4995              2      4995             3.9                  2.53
4996              1      4996             4.3                  1.53
4997              1      4997             3.2                  3.99
4998              7      4998             2.5                  2.13
4999              1      4999             1.9                  2.99

      Recommendation Accuracy  VR Experience Quality  Satisfaction
0                          97                    4.5             3
1                          90                    4.5             3
2                          94                    4.7             3
3                          92                    4.7             3
4                          96                    4.8             4
...                       ...                    ...           ...
4995                      100                    4.4             4
4996                      100                    5.0             4
4997                       89                    4.4             3
4998                       98                    4.4             3
4999                       92                    4.8             3

[5000 rows x 14 columns]
```

Algorithm 4

Normalization

Purpose

- Convert numerical values into a common scale.

Formula

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Code and Output :

```
import pandas as pd

df = pd.get_dummies(df, columns=['Interests', 'Accessibility', 'Site Name', 'Route ID'])

print(df)
```

```
      Route ID_4992  Route ID_4993  Route ID_4994  Route ID_4995  \
0             False          False          False          False
1             False          False          False          False
2             False          False          False          False
3             False          False          False          False
4             False          False          False          False
...             ...            ...            ...            ...
4995          False          False          False           True
4996          False          False          False          False
4997          False          False          False          False
4998          False          False          False          False
4999          False          False          False          False

      Route ID_4996  Route ID_4997  Route ID_4998  Route ID_4999
0             False          False          False          False
1             False          False          False          False
2             False          False          False          False
3             False          False          False          False
4             False          False          False          False
...             ...            ...            ...            ...
4995          False          False          False          False
4996           True          False          False          False
4997          False           True          False          False
4998          False          False           True          False
4999          False          False          False           True

[5000 rows x 5102 columns]
```

RESULT

Thus, missing values, duplicate records, and outliers were identified and handled successfully. Data transformation techniques such as normalization and encoding were performed, and the dataset was prepared for further exploratory data analysis and machine learning applications.